The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer as an anonymized observation runtime: multi-band evidence → BandPIT/WMI → a point-in-time (PIT) *world bundle*, with hard refusal when support is thin. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. ≲0.23 under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: under a *frozen full-schema* stress gate (max WMI ≈ 0.093<0.2), Compiled thin-abstain is 1.0 (100-day and full OOS n=2000) while Ungated soft judgment is only ≈0.68 / ≈0.56; Raw over-trades. Secondary—handoff: the *scoped-archive production valve* opens on ≈61% OOS days; live LLMs abstain 1.0 on closed days and act on ≈89% of open days, while a no-LLM tilt rule remains nonempty (Sharpe 0.77). **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation → decision → execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

Failure modes. No feed → refuse (safe, useless as a data end). Thin price fragment → trade and lose (unsafe for any downstream stack). Disclosed but ungated world → read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

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Table 1: End-to-end quant stack; this paper owns only the data end.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

What we build. An anonymized runtime (Section 4): multi-band evidence → BandPIT/WMI → JSON world bundle for GPT /DeepSeek /GLM /Gemin class models, served through a read API a decision layer can call. "World model" means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

How we evaluate (dual protocol). **RQ1 (primary—seeing):** Are ready /missing /tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Under a *frozen full-schema stress gate* (WMI over all schema bands; never clears 0.2 on this panel), does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (`mom5`), Blind. **RQ3a (handoff):** Under the *scoped-archive production valve*, do live LLMs obey closed days and act when open? Does a transparent no-LLM rule remain callable? **RQ3b (content):** On the same vintaged tilts, does an ungated content probe beat momentum (ΔCE; LOBO)?

Contributions. (1) Data-end semantics: epistemic observations, compilation Π_t, world-quality index with band scope, and seeing vs. gating with a lag–missingness bound. (2) Anonymized PIT compile path: collectors→BandPIT→service surface (bundle /health /handoff reads)→world bundles; frozen four-arm harness. (3) Live validation: grounding ≈0.97; full-schema stress abstain 1.0; scoped production valve with live LLM obedience (closed abstain 1.0; open act ≈0.89) plus no-LLM tilt handoff (Sharpe 0.77; LOBO content share 1.0). Estimand: data-end legibility, valve reliability, and handoff usability.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream at the data end of the stack. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act. Our estimand is therefore orthogonal to memory routers and debate judges: without a typed PIT world, those stacks consume whatever fragment the tool dump provides.

Selective prediction and PIT.. Abstention can be optimal under noise [4, 5]; we attach refusal to *world* quality $q(W_t)$ via a runtime field, not classifier confidence alone (Prop. 3.8). Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness (Prop. 3.6). Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation-Layer Semantics

This section states the data-end theory as *interface semantics*: what counts as an observation, how compilation yields a world, and when quality forces refusal. It is not a pricing theorem and not a generative dynamics model. Systems realization is Section 4; empirical tests are Section 5.

Definition 3.1 (Epistemic observation). $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views into a consumer object.

Definition 3.3 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ with scalar quality $q(W_t)$; $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$. W_t is the handoff object to decision.

Definition 3.4 (World-quality index). Ready/limited/missing counts disclose coverage. $\mathrm{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty in $[0, 1]$; ACWMI multiplies a regime weight. Band scope selects the support of (B, U, H): *full schema* (stress / ceiling audit) vs. the *declared consumer archive* (production valve). Scoped WMI scores only declared bands; it is a contract-completeness index, not a lowered $q^\star$. The Compiled interface exposes `should_ai_abstain=`$\mathbf{1}\{q(W_t) < q^\star\}$ (and, under archive scope, incompleteness) with `thin_world` (production $q^\star = 0.2$ for scoped WMI).

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t) < q^\star$. Tilts without a gate are a dump; a gate without a readable bundle is only a switch. The two interfaces are separable: Ungated keeps W_t and numeric $q(W_t)$ but removes the hard boolean.

Assumption 3.5 (Bounded lag and increments). Band j has max observation lag Δ_j; latent increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 3.6 (Lag–missingness reconstruction bound). *Under Assumption 1, any compiler that uses only in-clock observations satisfies*

$$\|\widetilde{S}_t - S_t\| \;\le\; C_{\mathrm{del}}\bar{\delta}\sum_j \Delta_j + C_{\mathrm{noi}}\varepsilon_t + C_{\mathrm{miss}} m_t,$$

where m_t is disclosed missing mass and ε_t collects measurement noise. Operationally, BandPIT maps age to ready/limited/missing so m_t and lag terms are readable in W_t; compilation cannot erase delay, but it can refuse to pretend the bound is zero.

Proposition 3.7 (Compilation ≠ dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\mathrm{AI}}$: ungated stale evidence can raise volume while lowering honesty, tightening rather than loosening Prop. 3.6.*

Proposition 3.8 (World-conditional abstention). *Let $c_{\mathrm{act}}(W_t)$ be the expected cost of non-abstain actions and c_{abs} the abstain cost. If $c_{\mathrm{act}}(W_t) > c_{\mathrm{abs}}$ and c_{act} is nonincreasing in $q(W_t)$, the Bayes action is abstain on a lower set $\{W : q(W) < q^\star\}$ [4]. A hard runtime flag implements that set; soft text does not.*

Proposition 3.9 (LOBO content vs. gating). *Deleting band j changes a downstream value functional through a content channel and a gating channel. Under an ungated rule the gating channel is shut, so the value gap identifies nonempty fields.*

Theory → protocol. Prop. 3.6–3.7 ⇒ RQ1 (recover disclosed readiness/tilts). Prop. 3.8 ⇒ RQ2 (hard flag vs. soft judgment under the full-schema stress gate). Def. 3.4 (band scope) + Prop. 3.8 ⇒ RQ3a (scoped production valve opens; handoff acts). Prop. 3.9 ⇒ RQ3b (ungated content / LOBO share).

Remark 3.10 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface—the data end of the stack.*

4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path (Figure 1, Table 2). Decision and execution sit downstream; this paper does not claim them. Paths and product names are omitted for double-blind review.

4.1 Layered design

(1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
(2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
(3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty/missingness gates implement Π_t; WMI/ACWMI and availability shocks O_t (band outage events for quasi-exogenous levers) are first-class.
(4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle reads, `/bundle/{symbol}/handoff` (valve-gated tilt action, no LLM), health exposing `should_ai_abstain` with `band_scope`, a paper world-model time slice, and availability-shock queries.
(5) **Consumer layer.** Frozen-prompt LLM/rule agents call an OpenAI-compatible adapter at temperature 0 under four arms.

4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of ≈3990 asset-days (2025-07 to 2026-08): exchange ready share ≈0.80 panel-wide (≈0.61 OOS), macro 1.0, alternative 1.0. **Full-schema stress WMI** scores permanently empty non-archive slots and caps at max ≈0.093, so a 0.2 gate never opens—the frozen LLM-arm protocol

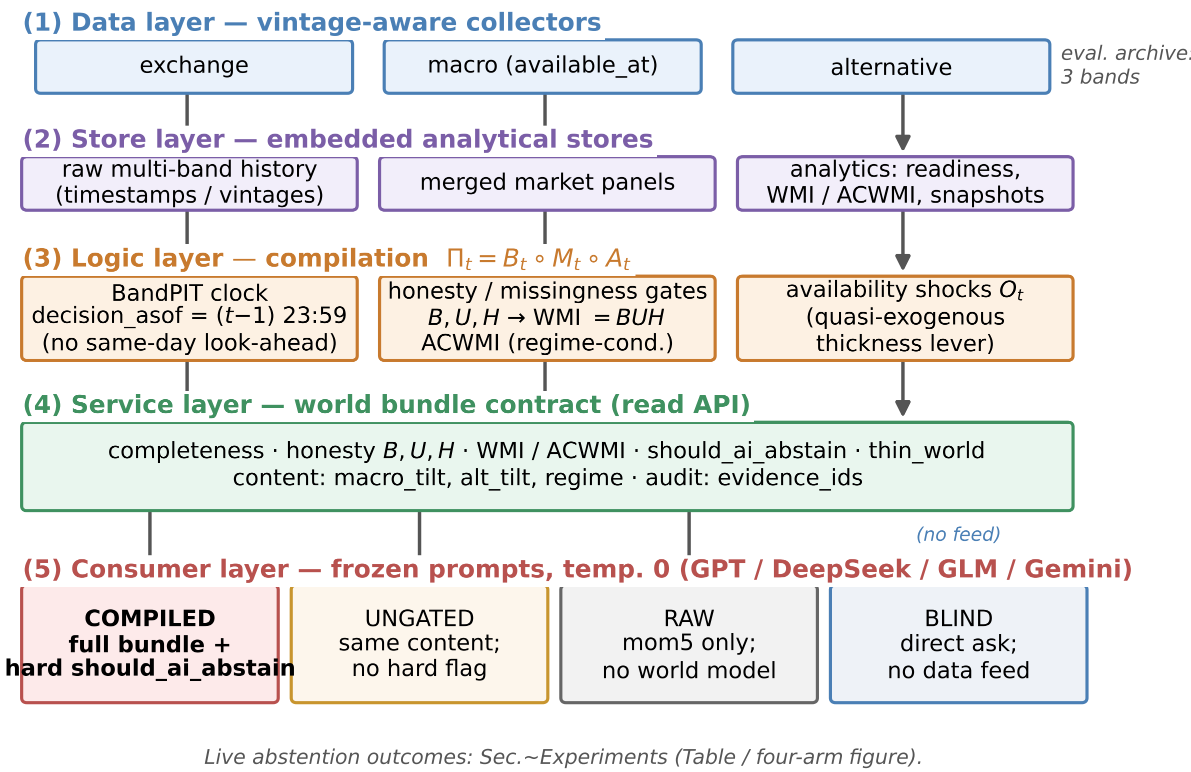

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \longrightarrow$ WMI; band scope; abstain; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / handoff / availability reads
Consumer harness	Four-arm protocol; temp. 0

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t, evidence IDs

(RQ2). **Scoped-archive production WMI** renormalizes B, U, H to $\{$exchange, macro, alternative$\}$ and opens iff the archive is complete and WMI ≥ 0.2 (RQ3a); this measures contract readiness, not a relaxed threshold.

4.3 PIT clock and freshness

For calendar day t, BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the bundle records `decision_asof`. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.6). Breadth, stability, and honesty aggregate into $\mathrm{WMI}_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The *scoped production valve* sets `should_ai_abstain` when the archive is incomplete or scoped WMI < 0.2, and exposes `thin_world`; bundles also disclose `full_schema_wmi` as a ceiling audit. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: same panel, two scopes—full-schema stress WMI stays under 0.2; scoped production WMI clears 0.2 on archive-complete days (shaded). Table 5: scoped production valve opens on ≈61% OOS days (≡ 3/3 ready); full-schema stress gate stays closed.

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

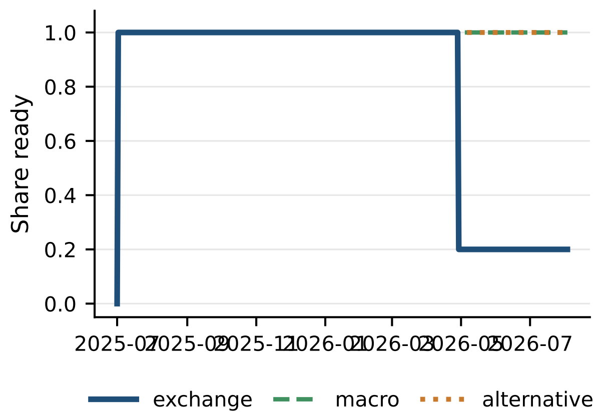

Figure 2: PIT readiness: what is visible in the archive over time.

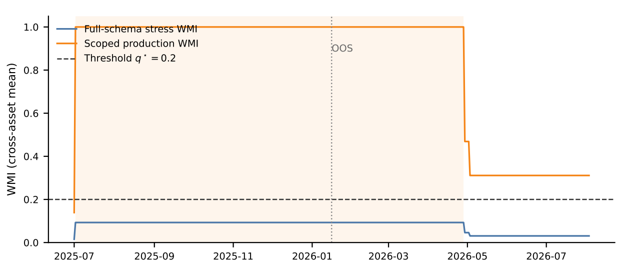

Figure 3: Dual-scope WMI on the same panel: stress ceiling vs. production contract (shaded = archive-complete).

Table 5: Dual gates vs. open share (OOS). Production = scoped; stress = full schema.

Gate	Open share	Note
Scoped production 0.2	0.61	complete ∧ WMI≥ 0.2
Full-schema stress 0.2	0.00	max WMI ≈0.093

tilts, and `evidence_ids`. Listing 1: exchange-gap day (scoped valve closed). Listing 2: 3/3-ready day where scoped WMI=1.0 opens the production valve; `full_schema_wmi`≈0.093 is disclosed as ceiling audit—same day, two scopes, not a lowered $q^\star$.

Listing 1: Exchange-gap bundle (Compiled; scoped closed).

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H; main-view / stale gates
Quality	WMI, ACWMI, band_scope, abstain, `thin_world`
Content	macro_tilt, alt_tilt, mom5
Audit	`evidence_ids`; EAR; `full_schema_wmi`

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1},
 "world_model_index":{"band_scope":"eval_archive",
   "wmi":0.139,"archive_complete":false,
   "should_ai_abstain":true,"thin_world":true,
   "full_schema_wmi":0.015},
 "macro_tilt":1.0,"alt_tilt":-1.0}
```

Listing 2: 3/3-ready day: scoped valve opens.

```
{"date":"2026-01-16","asset":"ADA",
 "completeness":{"n_ready":3,"n_missing":0},
 "world_model_index":{"band_scope":"eval_archive",
   "wmi":1.0,"archive_complete":true,
   "should_ai_abstain":false,"thin_world":false,
   "full_schema_wmi":0.093},
 "band_status":{"exchange":"ready","macro":"ready",
   "alternative":"ready"},
 "macro_tilt":1.0,"alt_tilt":-1.0}
```

4.5 Four arms and frozen prompts

Actions ∈ {bullish, bearish, neutral, abstain}. **Compiled:** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true (and when the world is judged thin). **Ungated:** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness. **Raw:** mom5 only—thin price wiring for the ladder. **Blind:** "Today is d. How should I trade a?" with no feed. **Dual protocol.** RQ1–RQ2 freeze *full-schema* bundles (max WMI ≈ 0.093), so Compiled abstains every OOS day—a closed stress test of the hard flag. RQ3a switches the same vendors to *scoped production* bundles: the valve opens on archive-complete days; we measure closed-day obedience, open-day action, and a parallel no-LLM tilt rule. Compiled vs. Ungated differ only in enforcement (Listings 3–4); prompts, IS/OOS cut, and seeds are frozen. On thin Compiled days the evidence-alignment proxy (EAR) is 1.0 across vendors: rationales stay bound to disclosed `evidence_ids`.

Listing 3: Compiled (excerpt): hard valve.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must match evidence_ids.
```

Listing 4: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI / missingness makes action
   unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why the hard valve is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the data end's job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.8). Ungated measures soft judgment from the same disclosed bundle; Raw measures fragment wiring; Blind measures no feed. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the valve.

4.6 Systems validation (engines)

Paper-lab smoke wires BandPIT/WMI/ACWMI and checks the dual gate: scoped archive opens on 3/3-ready statuses while full-schema stays closed; a handoff rule acts only when the scoped valve is open. Availability shocks O_t are queryable on the service surface.

5 Experiments

5.1 Setup

PIT panel ~399 days × 10 liquid names (≈3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses n=2000; no-LLM probes use ≈200 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain under the *full-schema stress gate* (CE as avoided loss, not alpha); RQ3a scoped production-valve LLM obedience plus no-LLM tilt handoff; RQ3b ungated tilt ΔCE vs. momentum.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility ≠ sufficiency chat. Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers "enough?" correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of "is this enough to trade?" is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) under the *full-schema stress* Compiled arm abstains on `should_ai_abstain`; Raw goes bullish from mom5 = +1 alone.

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

	Ready F1		Missing F1		Tilt-sign	
Model	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

Table 9: Thin abstention on identical 100 OOS days under full-schema stress (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. All live LLM arms in this subsection freeze *full-schema* bundles, so every OOS day is thin under the stress gate 0.2. Table 9 and Figure 4: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean ΔCE (Compiled−Raw) +1.07 is avoided loss under the stress valve, not an alpha claim.

Full-OOS replication. Table 11: on all 2000 OOS asset-days (all thin under the *full-schema stress* gate 0.2), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈0.56; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor):

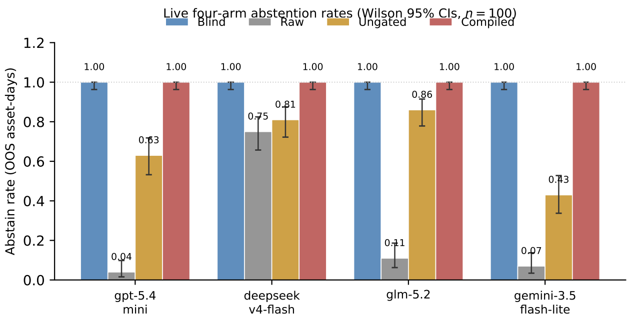

Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.

Table 10: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 11: Full-OOS thin abstention (n=2000; all thin under full-schema stress gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈0.56	–

*GLM coverage incomplete (Ungated n=1402; Raw n=675).

the failure mode that needs a data end is not "models never abstain," but "thin fragments make them act."

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek −3.12, GLM −4.28, Gemini −1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

Open slice (still under full-schema LLM arms). Table 12: on the band-thick subset (3/3 ready ≡ full-schema WMI ≥ 0.05; ≈61% OOS; n≈1224) Ungated keeps compiled content without the hard boolean—mean abs. ≈0.48, CE positive for every vendor (mean ≈+0.29; bootstrap ≈+0.31, CIs include 0). Full-schema Compiled still abstains at 1.0 here; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). The same thick days are where the *scoped production valve* opens; live scoped LLM handoff is RQ3a.

Table 12: Open slice under full-schema LLM arms (3/3 ready; n≈1224). Soft-judgment lower bound.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM n=398; Ungated GLM n=932; others n=1224. Bootstrap mean CE ≈+0.31 (B=5, 999 reps; one-sided p≈0.21).

Table 13: RQ3a live LLMs under scoped production valve (70 open + 30 closed).

Model	Abs. C_{cl}	Abs. C_{op}	CE C_{op}	CE R_{op}	ΔCE_{cl}
gpt-5.4-mini	1.00	0.03	−4.00	+0.09	+0.49
deepseek-v4-flash	1.00	0.24	−5.22	−0.21	+0.17
glm-5.2	1.00	0.11	−4.31	−0.39	+3.06
gemini-3.5-flash-lite	1.00	0.07	−4.27	−0.46	+2.64
Mean	1.00	0.11	–	–	+1.59

Abs. $C_{cl/op}$: Compiled abstain on closed/open days. ΔCE_{cl} = avoided loss vs. Raw on closed days.

Table 14: RQ3a no-LLM handoff. Scoped production valve vs. full-schema stress.

Policy (reads bundle fields)	Sharpe	CE†	Abstain
Full-schema stress 0.2	0.000	0.000	1.00
Scoped production 0.2 + tilts	**0.772**	**+0.253**	0.39
Momentum (mom5)	0.101	−0.075	0.00
Buy-and-hold	−1.399	−0.987	0.00

†CE = ann. mean − $\frac{1}{2}$ ann. variance (valve-aware path).

5.4 RQ3a (handoff): Scoped production valve with live LLMs

Having established seeing (RQ1) and stress-gate refusal (RQ2), we ask whether the *production* valve is callable. The full-schema stress gate never clears 0.2, so it cannot open—by design in RQ2. Scoped production WMI opens on ≈61% OOS days (≡ 3/3 ready). We re-run the same four vendors on a stratified sample of 70 open + 30 closed OOS asset-days with scoped bundles (seed fixed; temp. 0).

Table 13: on closed days Compiled abstains 1.0 for every vendor (Ungated soft rates ≈0.87–0.97; Raw often keeps trading). On open days Compiled acts on a mean ≈89% of days—the valve opens and models consume the bundle. Open-day CE does *not* beat Raw (mean ΔCE≈−4.2): opening the data-end valve is not an LLM alpha claim; policy skill remains downstream. Closed-day refusal avoids Raw losses (mean ΔCE≈+1.6 vs. Raw on the closed slice).

A transparent no-LLM rule on the same vintaged tilts remains the clean nonempty-field handoff (Table 14): Sharpe 0.772 / CE +0.253 when the scoped valve is applied, versus momentum and buy-and-hold; full-schema stress stays fully closed.

5.5 RQ3b (content): Nonempty fields

Table 15 and Figure 5: ungated always-on tilts vs. momentum on the same vintaged fields (ΔCE=0.334, p=0.034; LOBO content share

Table 15: RQ3b ungated content probe (no valve). Bootstrap ΔCE 0.334**.**

Policy	Sharpe	CE‡
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	**0.764**	**+0.130**
Tilts − Momentum	—	0.334 (p=0.034)

‡Content-harness CE (always-on; not comparable 1:1 to †).

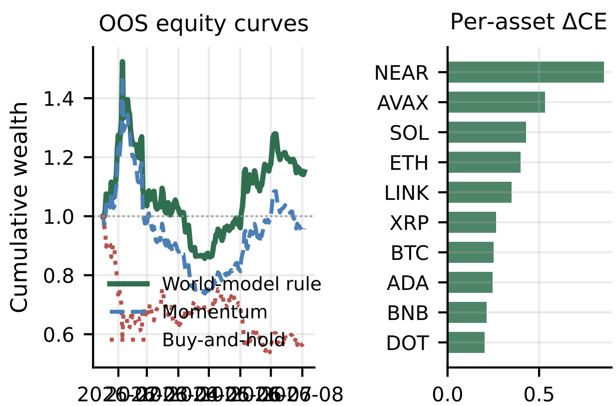

Figure 5: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.

Table 16: RQ3b content probe by archive density (OOS; ungated).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048§	0.129

*Block-bootstrap ΔCE; raw mean gap 0.336. §Always-on selection artifact.

Table 17: LOBO decomposition (ungated content probe). Content share 1.0**.**

Drop	ΔCE content	p	Content share
macro	−0.334	0.034	1.0
alternative	−0.334	0.034	1.0
exchange	0.000	1.000	—

1.0). This probe shuts the gating channel (Prop. 3.9); it is not the production valve.

Density split and LOBO channel. Table 16: open/thick ΔCE=0.315; on exchange-gap days the always-on probe still trades (high slice CE is a selection artifact, not a valve claim; ΔCE=0.129). Table 17 and Figure 6 (Prop. 3.9): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).

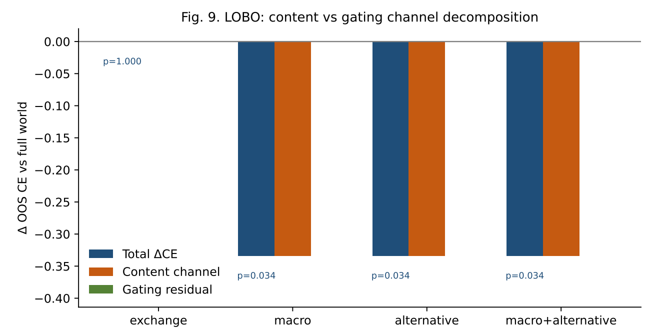

Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.

6 Discussion

What carries the paper. (i) The data end is legible: grounding ≈0.97, with sufficiency chat a poor substitute (RQ1 / Props. 3.6–3.7). (ii) Under the full-schema stress gate, soft text fails at scale; Blind refuses with no feed; Raw acts on fragments and loses; the hard flag works (RQ2 / Prop. 3.8). (iii) Scoped production WMI opens on band-thick days; live LLMs abstain 1.0 when closed and act when open, while open-day CE does not beat Raw—observation usability ≠ policy alpha (RQ3a / Def. 3.4+Prop. 3.8); a no-LLM tilt rule and LOBO content share 1.0 keep fields nonempty (RQ3b / Prop. 3.9). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

Architecture. Semantics (Section 3) define the interfaces; Runtime (Section 4) implements Π_t and the service handoff; Experiments (Section 5) test seeing, stress gating, then scoped handoff and content in that order. Trading PnL is intentionally secondary: it confounds observation quality with policy skill.

Design. World bundle not feature dump; previous-close PIT with explicit freshness; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Dual WMI scope separates stress audit from production readiness; health/bundle/handoff reads make both callable.

Reproducibility. Paper-lab smoke checks BandPIT/WMI/ACWMI and scoped-vs-full gates; PIT rebuild and the frozen four-arm harness regenerate the panel and live transcripts used here. Prompts, IS/OOS cut, seeds, and vendor IDs are fixed; credentials stay in environment variables. An anonymized harness ships on acceptance.

Estimand and scope. This paper measures data-end legibility, stress-gate reliability, and whether a scoped production valve is obeyed and callable. RQ1–RQ2 isolate seeing and hard refusal under the full-schema stress protocol; RQ3a switches to scoped production bundles for live LLM obedience and a parallel no-LLM tilt rule. Open-day LLM CE is reported as a usability check, not an alpha claim—policy skill sits downstream of the data end.

7 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation layer: compile asynchronous evidence into a consumable PIT world bundle, disclose structural missingness, and hard-refuse when the view is too thin. Grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. Raw collapse; sufficiency chat alone fails; the full-schema stress gate abstains 1.0 where soft judgment does not; under the scoped production valve live LLMs abstain 1.0 on closed days and act when open, while a no-LLM tilt rule stays nonempty (Sharpe 0.77). Build the quant data end first: compile so models can see; refuse when they must not act.

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